

ANESTHETICS

INTRODUCTION

- Any agent that produces a local or general loss of sensation, including pain.

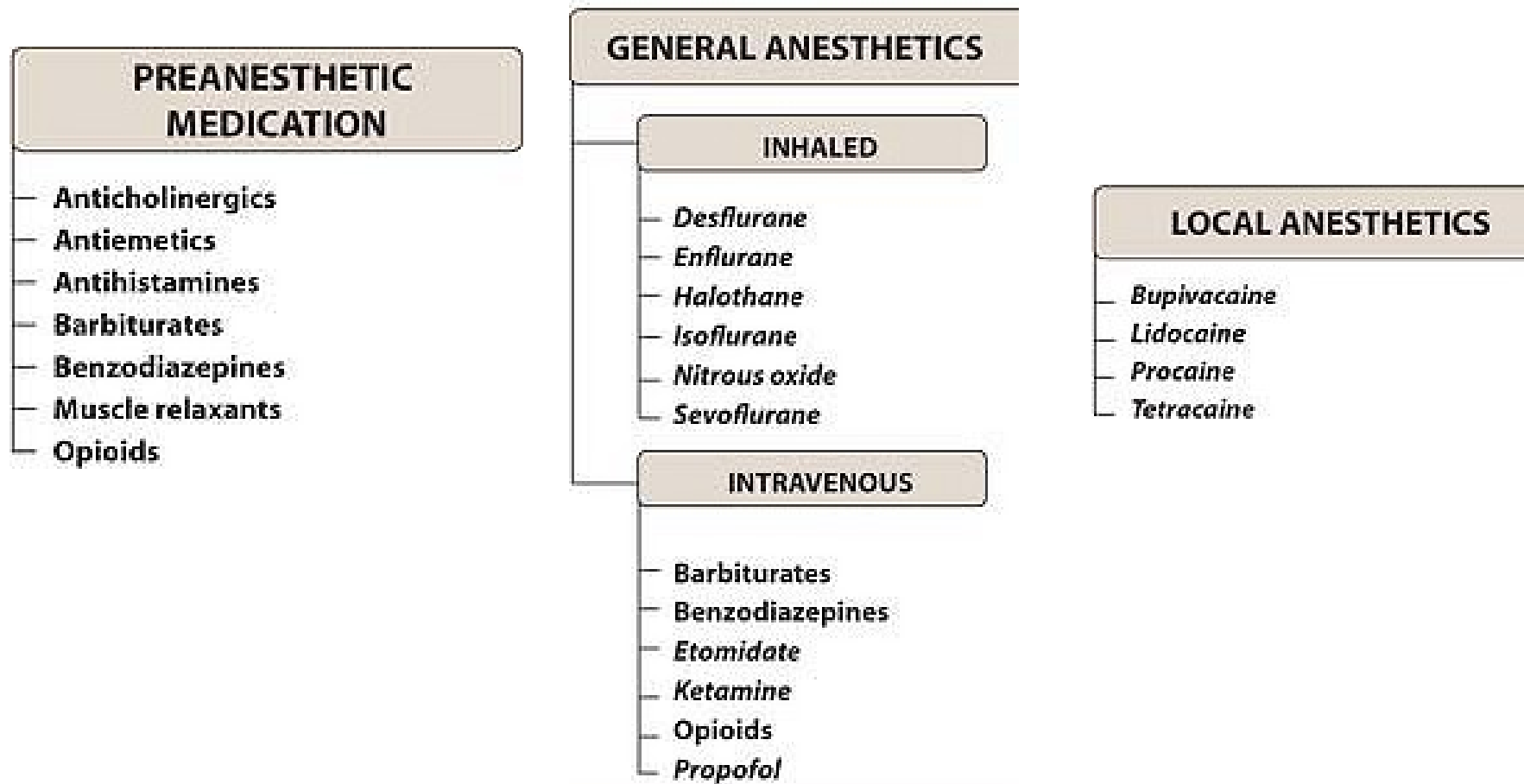
General anesthetics

- depress the CNS to a sufficient degree to permit the performance of **surgery and unpleasant procedures**.
- have **low therapeutic indices** and thus require great care in administration.

Local anesthetics (LAs) are **used to block transmission of impulses in nerve fibers**, to reduce or eliminate sensation.



CLASSIFICATION OF ANESTHETICS



- The components of the anesthetic state include

Loss of all sensation, especially pain

Sleep (unconsciousness) and amnesia

Abolition of somatic and autonomic reflexes

Immobility and muscle relaxation



STAGES OF GENERAL ANAESTHESIA

Stage-1

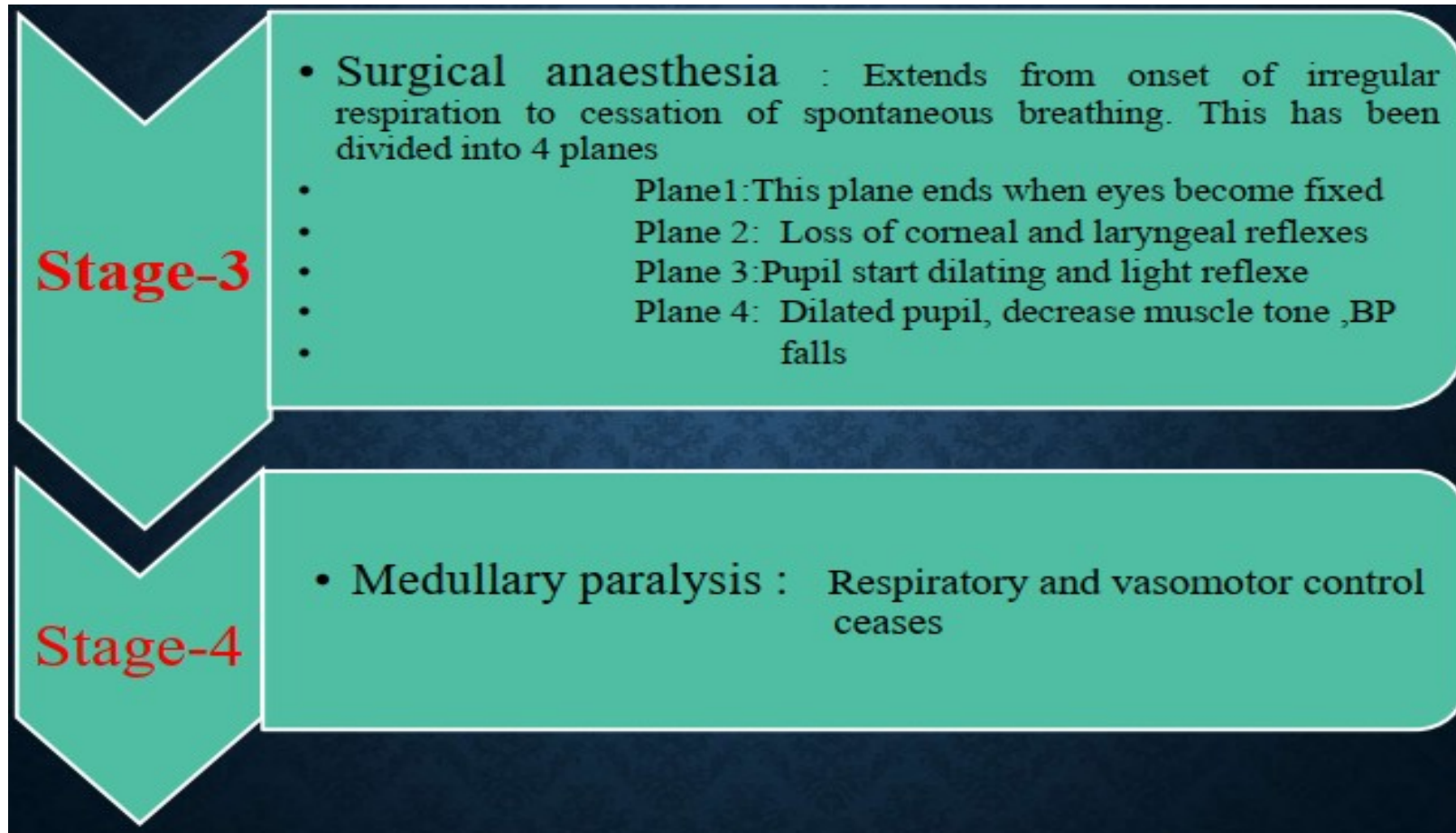
- **Analgesia** : Start from beginning of anaesthesia administration and last upto loss of consciousness, feels a dream like state, reflexes and respiration remains normal

Stage-2

- **Stage of delirium** : From loss of consciousness to beginning of irregular respiration. Apparent excitement is seen. Muscle tone increases. Jaws are tightly closed. Heart rate and blood pressure may rise.



STAGES OF GENERAL...



- The potency of general anesthetic agents is measured by determining the concentration of general anesthetic that prevents movement in response to **surgical stimulation**.
- For inhalational anesthetics, anesthetic potency is measured in **MAC units**.
- 1 MAC defined as the **minimum alveolar concentration** that prevents movement in response to surgical stimulation in 50% of subjects.



- Generally, the potency of intravenous agents
 - is defined as the **free plasma concentration** (at equilibrium) that produces **loss of response to surgical incision** (or other end points) in 50% of subjects.



MOLECULAR MECHANISMS OF GENERAL ANESTHETICS

- Most intravenous general anesthetics act predominantly through
- **GABAA receptors**
 - chloride channels gated by the inhibitory GABAA receptors are sensitive to a wide variety of anesthetics.
 - Modulating GABA receptor
- **Glycine receptor**
 - Glycine-gated Cl⁻ channels may play a role in mediating inhibition by anesthetics
- **NMDA receptors**
 - glutamate-gated cation channels that are somewhat selective for Ca²⁺
- Also has effect on neuronal **nicotinic receptors** and **5-HT₃ receptors**



INHALATIONAL ANAESTHETICS

- Inhalational anaesthesia refers to the delivery of gases or vapours to the respiratory system to produce anaesthesia

Pharmacokinetics

- ❖ both volatile and gaseous, are taken up through gas exchange in the alveoli of the lung.
- ❖ Uptake from the **alveoli** into the **blood** and **distribution** and partitioning into the effect compartments within the body are important determinants of the kinetics of these agents.

Factors affecting Uptake

- ❖ **Inspired concentration and ventilation-** The driving force for uptake is the ratio between **inspired** and **alveolar concentration**.

Solubility—of the most important factors influencing the transfer of an anesthetic from the lungs to the arterial blood is its solubility characteristics

- ✓ the **blood:gas partition coefficient** is a useful index of solubility and defines the relative affinity of an anesthetic for the blood compared to the affinity for inspired gas.
- ✓ **Desflurane** and **nitrous oxide**, are relatively insoluble in blood, display low partition coefficients.
- ✓ **Halothane** and **isoflurane**... are more soluble



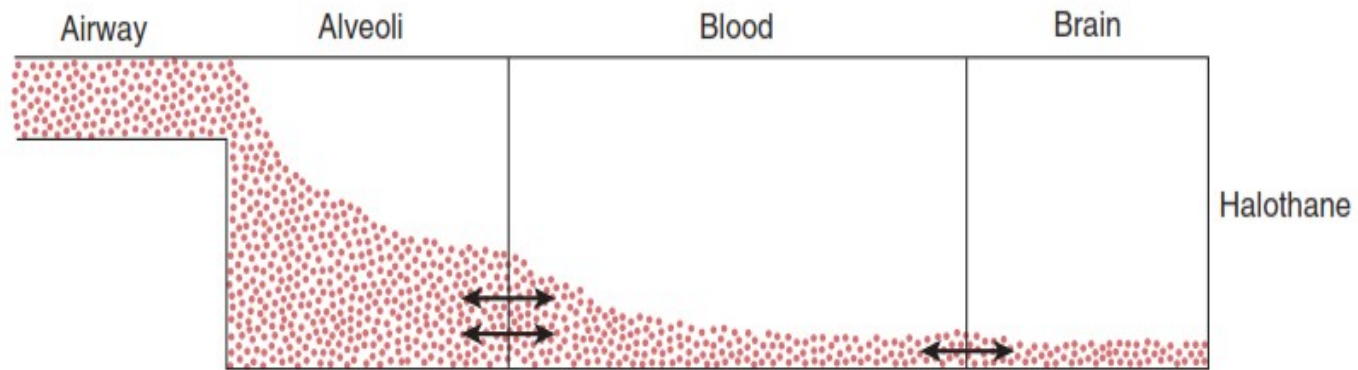
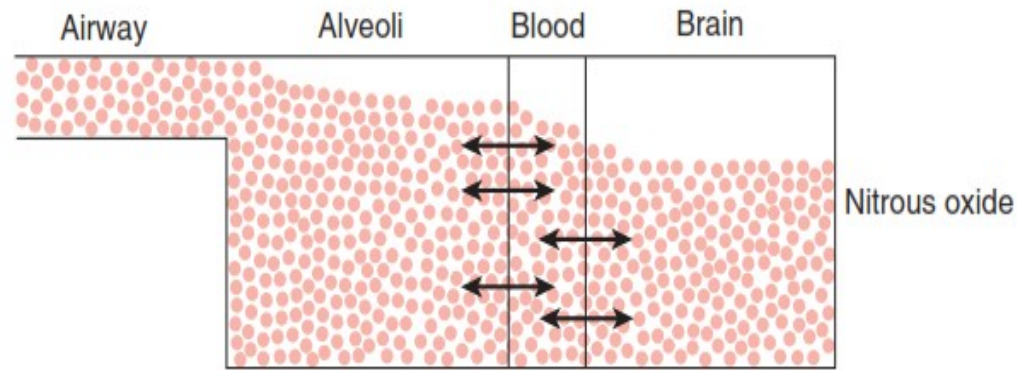


TABLE: PROPERTIES OF ANESTHETICS

Anesthetic ⇅	MAC (%) ⇅	Blood/Gas ⇅	Oil/gas ⇅	Brain/Blood ⇅	Muscle/Blood ⇅	Fat/Blood ⇅
Nitrous oxide	105	0.47	1.4	1.1	1.2	2.3
Halothane	0.74	2.4	224	2.9	3.5	60
Isoflurane	1.15	1.4	97	2.6	4.0	45
Sevoflurane	2	0.65	42	1.7	3.1	48
Desflurane	5.8	0.42	18.7	1.3	2.0	27

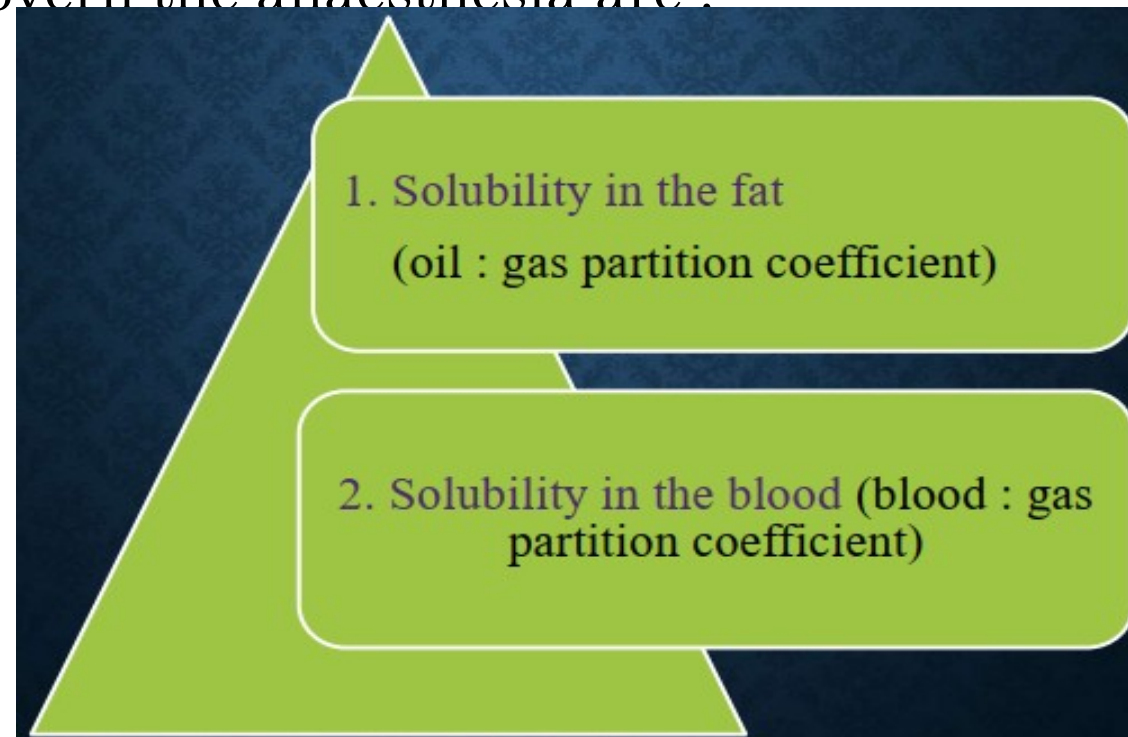
Cardiac output—Changes in the flow rate of blood through the lungs also affect the uptake of anesthetic gases from the alveolar space.

- An increase in pulmonary blood flow (ie, increased cardiac output) will increase the uptake of anesthetic, thereby slowing the rate of induction of anesthesia,
- The increased uptake of anesthetic into the blood caused by increased cardiac output will be distributed to all tissues.
- Since cerebral blood flow is well regulated, the increased anesthesia uptake caused by increased cardiac output will predominantly be distributed to tissues that are not involved in the site of action of the anesthetic.

MAC50?



- Two important characteristics of Inhalational anaesthetics which govern the anaesthesia are :



- **Oil-gas partition coefficients :**

- It indicates the amount of gas that is soluble in oil phase
- It is a measure of lipid solubility of anaesthetic
- It a measure of anaesthetic **potency**
- Higher the solubility of general anaesthetics in oil greater is the anaesthetics action

- **Blood-gas partition coefficients :**

- The ratio of the concentration in blood to the concentration in the gas phase
- Lower the blood : gas coefficient → **faster the induction and the faster the recovery**



Anaesthetic	Blood/Gas	Oil/gas
Nitrous oxide	0.47	1.4
Halothane	2.4	224
Isoflurane	1.4	97
Sevoflurane	.65	42
Desflurane	.42	18.7



Ether

- It is explosive
- **Irritant to respiratory tract**
- High incidence of **nausea and vomiting** during induction and post-surgical emergence



Nitrous oxide

- Colourless, odourless gas at room temperature.
- **Very insoluble in blood and other tissues** (quick recovery)
- **Rapid induction of anaesthesia** and rapid emergence following discontinuation of administration.
- **does not depress respiration, nor does it produce muscle relaxation.**
- Completely eliminated by the **lungs**.
- It is **weak anaesthetic** and **powerful analgesic**.
- The mac value is **105%**.
- Causes **megaloblastic anaemia**.
- Used as **adjunct** to supplement other inhalationals.



Halothane

- **Volatile liquid** at room temperature.
- Light sensitive
- **High fat solubility** => slow induction & recovery
- Eliminated unchanged **via lungs**
- Commonly used in **children**, where preoperative placement of an iv catheter can be difficult
- It is marketed in amber bottles with thymol added as a preservative
- Metabolized in liver by **Cyt-P450**



HALOTHANE ...

Advantages of halothane:

- pleasant smell, therefore is accepted by many patients
- Very potent anesthetic (MAC= 0.75%)
- Non-irritant
- Bloodless field for plastic surgery because of the fall in blood pressure.
- Solubility in blood is intermediate $\Rightarrow$ reasonable induction & recovery.
- No post operative nausea & vomiting
- No bronchospasm, no coughing, hence suitable for tracheal intubation.

Disadvantages of halothane:

- Little analgesic activity
- Little muscle relaxation
- Fall in blood pressure because of myocardial depression
- Expensive



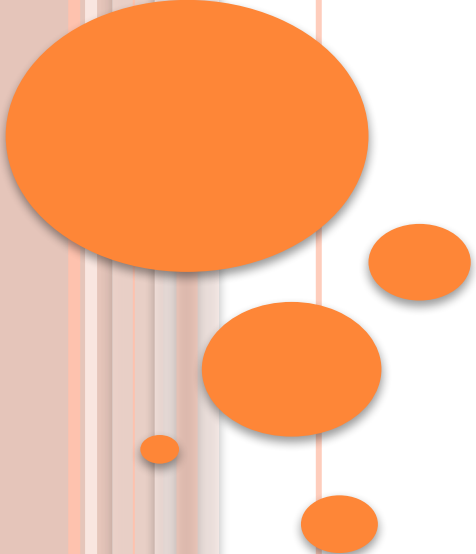
- **Side effects of halothane :**
 - CVS : Cardiac arrhythmia, depression of myocardial contraction.
 - Respiratory system : Depression of respiration
 - Muscles : Malignant hyperthermia
 - Kidney : Decrease renal blood flow and GFR
 - Liver and GIT: Cause halothane induced hepatitis & nausea and vomiting
- **DRUG INTERACTION :** Halothane + adrenaline, theophylline => arrhythmia may be precipitated.
- **CONTRAINDICATION :** Hepatic dysfunction and/or jaundice



Sevoflurane

- Pleasant smell , non irritant and **bronchodilation** makes it agent of choice for **paediatric anaesthesia**.
- 2nd agent of choice for
 - Neuro anaesthesia.
 - Cardiac anaesthesia .
 - Asthmatics.
- Agents that should not be given with soda lime.
 - Trielene.(trichloro ethylene)
 - Sevoflurane.
 - Desflurane





INTRAVENOUS ANAESTHETICS

THIOPENTAL

- **Ultrashort** acting barbiturate
- a potent anesthetic and a weak analgesic ⇒ **require a supplement of analgesic Admn**
- **High lipid solubility** rapid entry into the brain
- Rapid onset (20 sec) , short duration
- **Barbiturates** should not be administered to patients with acute intermittent **porphyria** (when the **body** cannot convert compounds called '**porphyrins**' into heme) because they increase the production of porphyrins.



THIOPENTAL CONT'D

- Risk of sever **vasospasm** if accidentally injected into artery
- Depress cerebral blood flow
- **Decrease intracranial pressure**
- Tissue **necrosis—gangrene**, hypotension, apnea
- Build-up in adipose tissue = very long emergence from anaesthesia

Methohexital has a shorter **elimination half-time** than thiopental

- Recovery from intravenous anesthetics is due to **redistribution** from sites in the CNS



Propofol

- **Rapid onset and have a short duration of action**
- Highly protein bound *in vivo* and is metabolised by conjugation in **the liver**
 - Onset of action= 40 sec.
 - Narcotics are required for analgesia
 - Replaced thiopental
 - Doesn't cause postanesthetic nausea and vomiting
- Useful for spinal surgeries
- Very good **anesthetic for induction and maintainance of anesthesia with no accumulation effect**
- Side-effects are pain on injection, hypotension and transient apnea following induction
- Used for the **induction, maintenance of GA and sedation**
- Useful for **day-case surgery**



Ketamine

○ Dissociative anaesthetic

- the patient is unconscious but appears to be awake and does not feel pain.
- provides sedation, amnesia, and immobility with or without loss of consciousness (hypnosis)
- the only intravenous anesthetic that possesses anesthetic and analgesic properties, as well as the ability to produce dose-related cardiovascular stimulation.

MOA

- NMDA Receptor Antagonist
- Cardiovascular stimulant
- Catatonia, analgesia, and amnesia without loss of consciousness
- Useful for anesthetizing **patients at risk for hypotension** and **bronchospasm** and for certain paediatric procedures



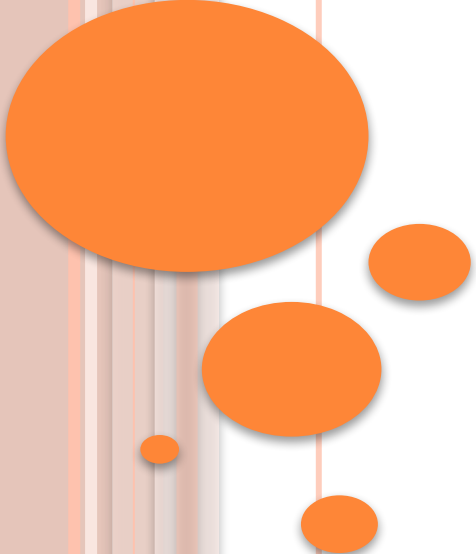
Etomidate

- A **hypnotic** agent but lacks **analgesic** activity

MOA : potentiation of GABAA-mediated chloride current, like most other intravenous anesthetics.

- Rapid but short-acting.
 - Only used for patients with **coronary artery** disease or cardiovascular dysfunction, such as shock (an alternative to propofol and **barbiturates**)
- Clearance of **etomidate** is about **five times faster** that of thiopental, as reflected by a shorter elimination half-time
- Minimal **change in cardiac function and respiratory rate**
- **Not analgesic**
- **Cause pain on injection and nausea postoperatively**
- Prolonged administration may cause **adrenal suppression**





LOCAL ANAESTHETICS

FEATURES OF LOCAL ANAESTHETICS

- Should **have quick onset of action**
- **Should not be irritating to skin & mucous membranes**
- Duration of action must be **long enough to allow desired surgery to be completed**
- Should be **effective on both injection & local application**
- Should have **low Systemic toxicity**



- Should **not cause any permanent damage** on any tissue.
- Should be relatively **free from producing allergic reaction**.
- Should **be stable in solution and readily undergo biotransformation**.

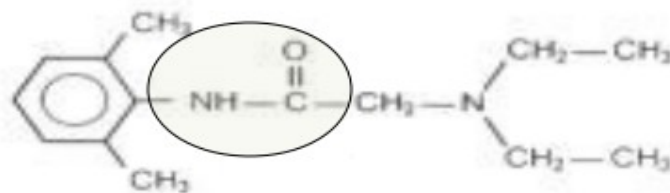


BASED ON LINKAGE THEY CAN BE CLASSIFIED AS

Amino-esters

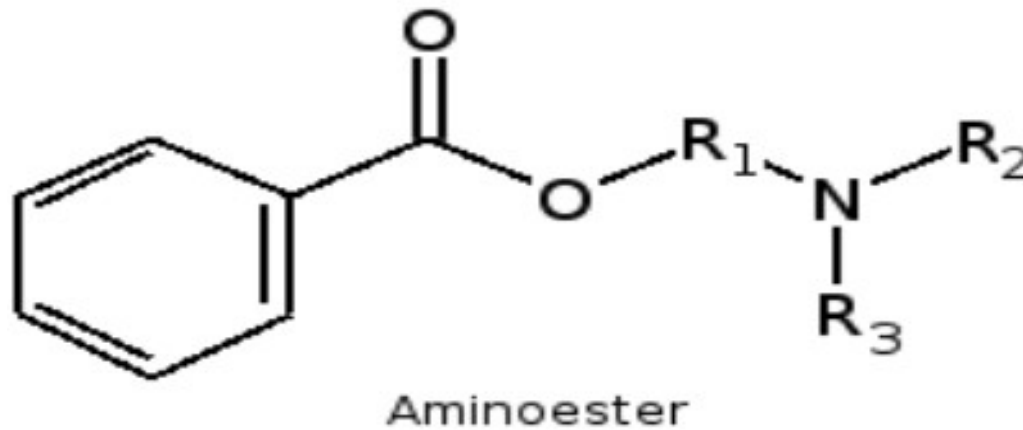


Amino-amides



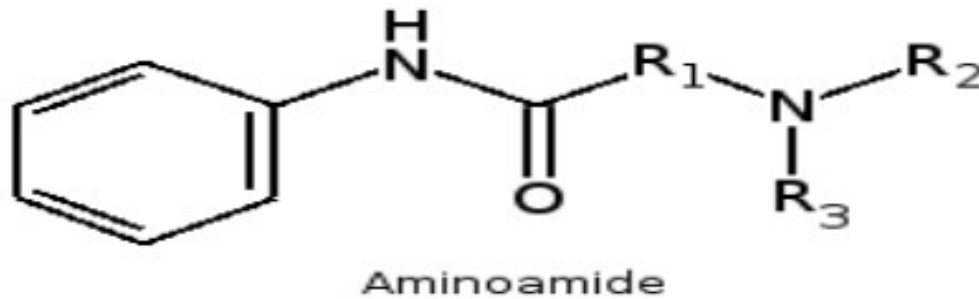
ESTERS:

- cocaine, procaine, tetracaine, and chlorprocaine.
- hydrolyzed in plasma by pseudo-cholinesterase.



AMIDES:

- lidocaine, mepivacaine, prilocaine, bupivacaine, and etidocaine.
- metabolized in the liver to inactive agents.
- True allergic reactions are rare (especially with lidocaine)



ADVANTAGE OF AMIDE LAs OVER ESTER LAs

- Produce more intense and longer lasting anaesthesia
- Bind to α 1 acid glycoprotein in plasma.
- Not hydrolysed by plasma esterases.
- Rarely causes hypersensitivity reaction.



LOCAL ANESTHETICS CLASSIFICATION

Injectables

A. Low potency & short duration

- **Procaine**
- Chloroprocaine

B. Intermediate potency & duration

- **Lignocaine (lidocaine)**
- Prilocaine

C. High potency & long duration

- **Tetracaine**
- **Bupivacaine**
- Ropivacaine
- Dibucaine

- **2. Surface Anesthetics:** -
Usually applied on mucous membrane or abraded skin Include.
 - Cocaine
 - Lignocaine
 - Tetracaine
 - Benoxinate
 - Benzocaine
 - Butyl amino benzoate
 - Oxethazaine



2. SURFACE ANAESTHETIC:

○ SOLUBLE

cocaine

lidocaine

tetracaine

benoxinate

INSOLUBLE

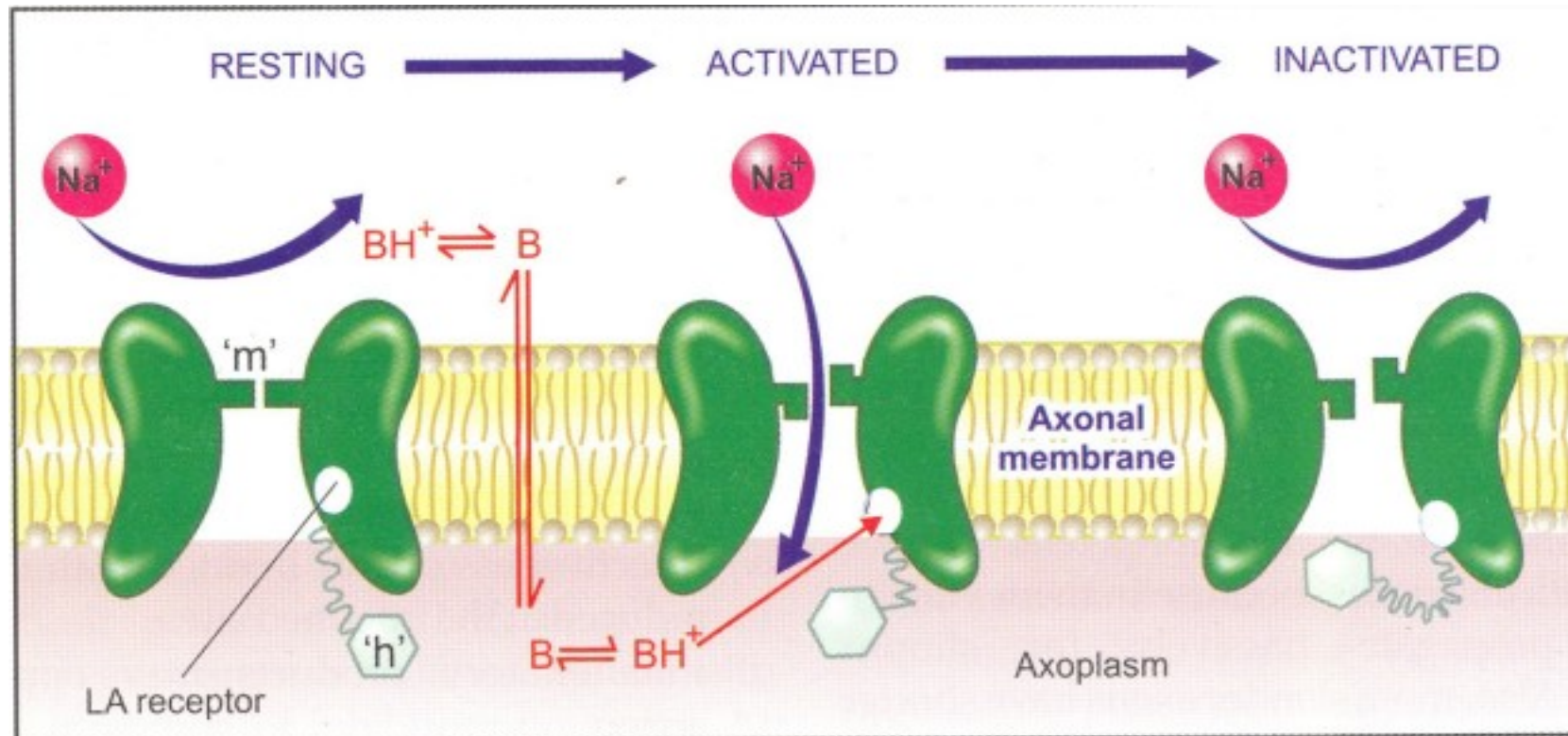
benzocaine

butylaminobenzoate

oxethazaine



MECHANISM OF ACTION



- LA blocks the nerve conduction by reducing entry of Na^+ through the voltage gated channels
- Due to this, they block the initiation & propagation of nerve impulse.
- At higher doses it also blocks
 1. Voltage gated Ca^{2+} channels
 2. K^+ channels



VASOCONSTRICTOR

- **Vasoconstrictor** used – adrenaline (1:50,000 to 1:200,000).
- **prolongs duration of action of LAs by decreasing rate of removal from local site into circulation**
- **Enhances intensity of nerve block**
- **Reduces systemic toxicity of LAs**
- **Provides more bloodless field for surgery**



SYSTEMIC ACTIONS

- **CNS:** All LAs are capable of producing a sequence of stimulation followed by **depression**
- **CVS:** LAs are cardiac depressants, but no significant effects are observed at conventional doses. But at high doses they decrease automaticity, excitability, contractility, conductivity.
- **BLOOD VESSELS:** LAs tend to **produce fall in BP**. This is primarily due to **sympathetic blockade**, but higher doses cause direct relaxation of arteriolar smooth muscle



SYSTEMIC TOXICITIES OF LOCAL ANESTHETICS

- **Central Nervous System Toxicities:**

- Excitement: Tremors, shivering, and convulsions characterize the CNS excitement.
- Depression: respiratory depression at higher doses

- **CVS derangement**— High plasma titers **may depress the cardiovascular system directly.**

- Blood pressure may **fall because of arteriolar dilation, myocardial depression, and/or cardiac conduction disruption.**
- Treatment includes patient positioning, IV fluids, and vasopressors.



- **Haematological:** large dose of prilocaine $> 10\text{mg/Kg}$
Accumulation of its metabolite- o toluidine-oxidises
haemoglobin to methaemoglobin
- Treatment: I.V methylene blue /ascorbic acid-a reducing
agent – convert methaemoglobin to Hb.
- **Hypersensitivity:** Some patients are hypersensitive
(allergic) to some local anesthetics.
 - Ester type



PHARMACOKINETICS

Absorption

- Local anesthetics are absorbed when ingested.
- Some local anesthetics may be absorbed in toxic amounts after topical use.
- Absorption after an injection depends on drug solubility in lipid and in water, tissue vascularity and local anesthetic and vasoconstrictor effects on local circulation.



- **Distribution:** amides-wide distribution –I.V-lipophilics taken up by highly perfused organs-then moderately perfused
- **Ester** type- **short** plasma half life

Metabolism and excretion

- **Esters** are hydrolyzed **by plasma and liver esterases**. Longer-acting esters are often metabolized more **slowly**..
- Patients with altered pseudo-cholinesterase activity may be highly sensitive to these drugs.
- **Amides** are metabolized **in the liver by cyp450**.-Ndealkylation then hydrolysis except prilocaine hydrolysis first-o **toludine-can** cause **methhamoglobinemia**



PRECAUTIONS AND INTERACTIONS

- Inject the **LA** slowly & **take** care not to exceed the maximum safe dose, especially in children.
- **Propranolol** may reduce metabolism of lidocaine and other amide LAs by reducing hepatic blood flow.
- **Vasoconstrictor** (Adr) containing LA should be avoided for patients with **ischemic** heart disease, cardiac arrhythmia, uncontrolled hypertension those receiving β -blockers or tricyclic antidepressants



COMMONLY USED LOCAL ANESTHETICS

Cocaine

- Current clinical use of cocaine is largely restricted to **topical anesthesia** for ear, nose, and throat procedures,
- vasoconstriction can serve to reduce bleeding.
- has diminished in favor of because of concerns about
 - ❖ Systemic toxicity, as well as the
 - ❖ inconvenience of **dispensing** and **handling** this controlled substance.

Articaine

- unique among the amino-amide anesthetics in having a **thiophene** and additional **ester group** (promote for hydrolysis)
- more **effective**, and **safer**, than **lidocaine**,
- widespread popularity in **dental anesthesia**

Benzocaine

- benzocaine's pronounced **lipophilicity** has **diminished** its use to topical anesthesia.
- induce **methemoglobinemia**recently diminished its use.

BUPIVACAINE

- ❖ Due to **cardiotoxicity**, bupivacaine is often avoided for techniques that demand high volumes of concentrated **anesthetic**, such as **epidural or peripheral nerve blocks** performed for surgical anesthesia.
- ❖ relatively low concentrations ($\leq 0.25\%$) are frequently used to achieve prolonged peripheral anesthesia and analgesia for **postoperative pain control**,

CHLOROPROCAINE

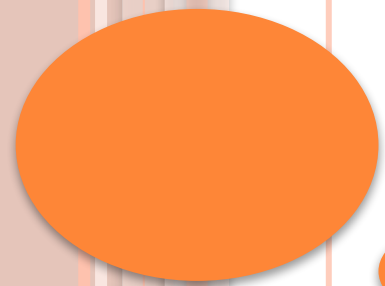
- ❖ Widely used in **obstetrical anesthesia** where its rapid
 - 🌱 hydrolysis served to minimize risk of systemic toxicity or fetal exposure
- ❖ The **onset and duration of action** of spinal **chloroprocaine** are shorter than those of **lidocaine**
- ❖ Is used in a situation where indwelling catheter is required, such as **caesarian section** for a **laboring parturient** with a compromised fetus.

- ❖ Structurally similar to **bupivacaine** and **ropivacaine**
- ❖ Mepivacaine displays clinical properties that are comparable to **lidocaine** but it is **vasoconstrictor**
- ❖ Lidocaine has retained its dominance over mepivacaine for **epidural anesthesia**, where the routine placement of a **catheter negates** the importance of a longer duration.
- ❖ Mepivacaine is slowly metabolized by the fetus, making it a poor choice for epidural anesthesia in the **parturient**.

CLINICAL USES

Method of Administration	Uses	Drugs
Surface anesthesia	Nose, Mouse, Urinary tract	Lidocain
Infiltration anesthesia	Direct injection in to tissues to reach nerve branches and terminals. (Minor surgery)	Most
Regional anesthesia	Local anesthetic injected IV distal to pressure cuff, limb surgery	Mainly lidocaine
Nerve block anesthesia	LA injected close to nerve trunks. Dentistry	Most
Spinal anesthesia	LA injected in to subarachnoid space. Pelvis surgery	Mainly Lidocaine
Epidural anesthesia	LA injected into epidural space. (Labour)	Mainly lidocaine





OPIOIDS

SOURCE

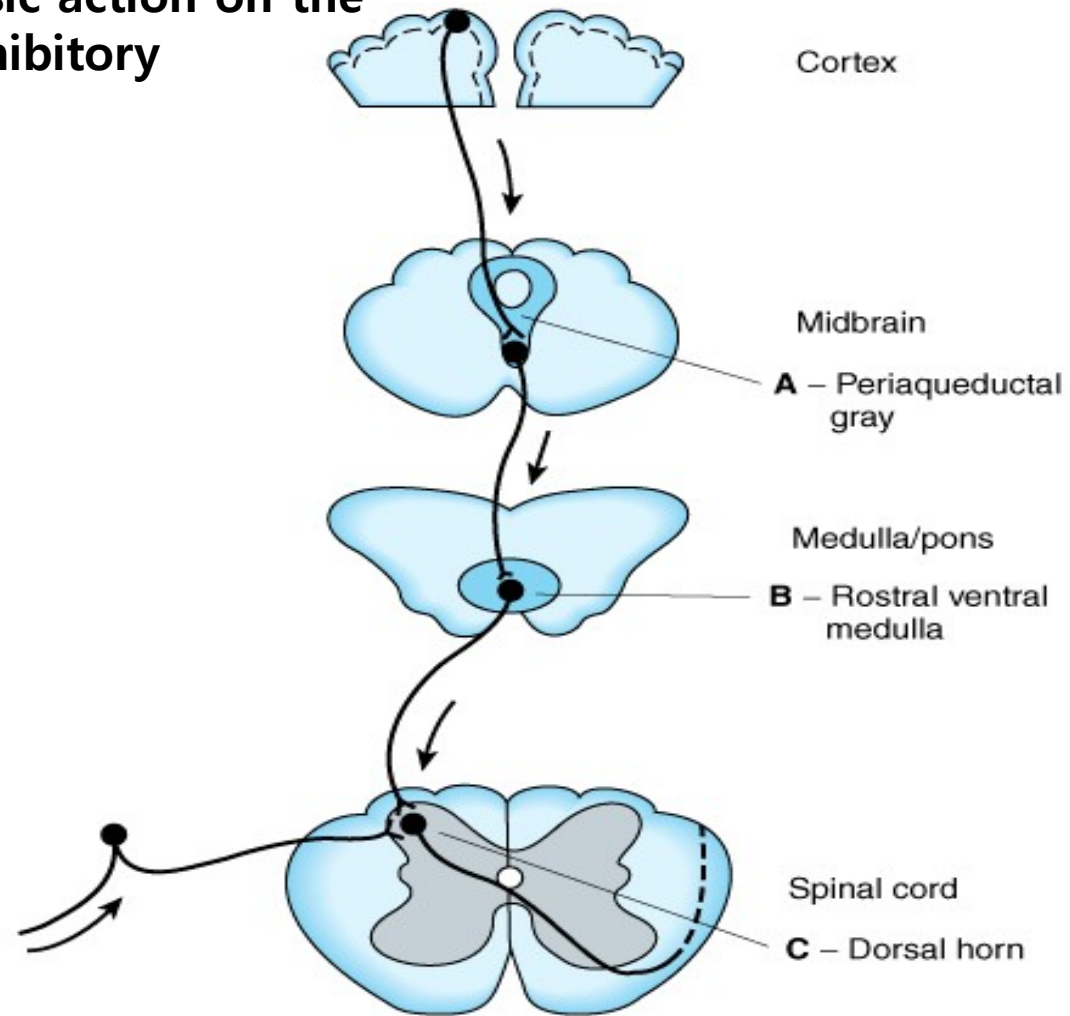
- Opium, the source of morphine, is obtained from the poppy, *Papaver somniferum* and *P. album*.
- Opium contains many alkaloids, the principal one being morphine, which is present in a concentration of about 10%.
- Codeine can also be found in opium and is synthesized commercially from morphine.



- The term opioid describes all compounds that work at opioid receptors.
- The term opiate specifically describes the naturally occurring alkaloids: morphine, codeine, thebaine, and papaverine.



Opioid analgesic action on the descending inhibitory pathway



OPIOID RECEPTORS

- Mainly 3 (three) types of receptors – μ (mu), κ (kappa) and δ (delta)
- **Subtypes:** μ_1 , μ_2 , κ_1 , κ_2 , κ_3 , δ_1 and δ_2
- **Location:** Peripheral Nerve endings, in spinal cord, Periaqueductal gray (PAG) in midbrain and Brain stem (medulla, hypothalamus and also amygdala)
- Opioids are – agonists, partial agonist or competitive antagonists of these receptors
- Overall effect depends on nature of interaction and affinity to these
- Morphine is agonist of all but affinity is higher for mu



	μ receptor	κ receptor	δ receptor ¹
Location	$\mu 1$ - supraspinal $\mu 2$ - spinal	$\kappa 1$ - spinal $\kappa 3$ -supraspinal	Spinal supraspinal
Effects	Analgesia Respiratory depression Sedation Euphoria Miosis Physical dependence	Spinal analgesia Dysphoria Sedation Psychomimetic Physical dependence	Spinal analgesia Affective behaviour (Supraspinal) Respiratory depression Reduced GI motility



OPIOID RECEPTOR INTRACELLULAR MECHANISM

- All are **G-protein coupled receptors**
- Located on presynaptic neurones
- **Inhibits release** of transmitters – NE, DA, 5-HT, GABA and Glutamate
- Activation of $\mu/\kappa/\delta$ $\downarrow$ cAMP formation - Opening of K^+ channel $\rightarrow$ suppression of N type of Ca^{++} channels
- Ultimately Hyperpolarization and reduced intracellular Ca^{++} $\rightarrow$ Reduced Neurotransmitter release



ENDOGENOUS OPIOID PEPTIDES

○ Endorphins:

- Derived from POMC (**Proopiomelanocortin**)
- β -endorphins: 2 Types - β -endorphin1 and β -endorphin-2
- Primarily μ agonist and also has δ action

○ Enkephalins:

- Derive from Proenkephalin
- Met-ENK and leu-ENK
- Met-ENK - Primarily μ and δ agonist and leu-ENK – δ agonist

○ Dynorphins:

- Derive from Prodynorphin: DYN-A and DYN-B
- Potent κ agonist and also have μ and δ action



PHARMACOLOGICAL ACTIONS

Sedation:

- Drowsiness and indifference to surroundings
- Inability to concentrate and extravagant imagination colorful day dream
- Apparent excitement
- Larger doses produce sleep – EEG resembles normal sleep

Mood effects:

- Euphoria
- Dysphoria



Depression:

- Respiratory center depression –
Both rate and depth of respiration are diminished
 - Dangerous in Head injury and asthmatics
- Cough Centre –Depressed
- Temperature regulating center – depressed
- Vasomotor center – high doses

Stimulation:

- CTZ – sensitize CTZ to vestibular and other impulses
- Miosis
- Vagal centre –Bradycardia
- Hippocampal cells – convulsions (inhibition of GABA release)



Neuro-endocrine:

- GnRH and CRH are inhibited – FSH, LH and ACTH levels are lowered
- Decrease in levels of Sex hormone and corticosteroids, but no infertility
- Increases ADH release – oliguria

CVS: NO DIRECT EFFECT ON HEART

- Vasodilatation – histamine release, depression of vasomotor centre and directly on blood vessels decreasing the tone
- Cardiac work reduction due to consistent vasodilatation



GIT: CONSTIPATION

- Due to direct action on intestine reducing propulsive movement, spasm of sphincters, decrease in all GIT secretions

Smooth Muscles:

- Biliary Tract: Biliary colic – closure of sph. Of Oddi
- Bladder: Urinary urgency but difficulty
- Bronchi - Bronchospasm



ADVERSE EFFECTS

- Respiratory Depression: Infant and Old
- Vomiting
- Sedation, Mental Clouding – sometimes dysphoria
- Hypotensive effect
- Rise in Intracranial Pressure
- Apnea: Newborn
- Urinary retention



OPIOIDS - CLASSIFICATION

1. Natural Opium Alkaloids: Morphine and Codeine
2. Semi-synthetic: Diacetylmorphine (Heroin) and Pholcodeine
3. Synthetic Opioids:
 - **Phenylpiperidines:**
 - Pethidine (Mepiridine) and its congeners – Diphenoxylate and Loperamide
 - Fentanyl and its congeners – sufentanil, remifentanil and alfentanil
 - **Phenyl-heptylmines:** Methadone and congeners like Propoxyphene and Dextropropoxyphene
 - **Benzomorphans:** Pentazocine
 - **Morphinan** compounds and congeners: Levorphanol and Butorphanol



Fentanyl

- 80-100 times more potent than morphine
- few cardiovascular effects
- little tendency to release histamine.
- Because of high lipid solubility, it enters brain rapidly and produces **peak analgesia in 5 min** after i. v. injection.
- The **duration of action is short**: starts wearing off after 30-40 min due to redistribution
- **Transdermal fentanyl** has become available for use in cancer



PETHIDINE

- Morphine Vs Pethidine:
 - 1/10th as potent as Morphine, but Efficacy is similar
 - Produces as much sedation, euphoria and respiratory depression in equianalgesic dose and similar abuse potential
 - Less spasmodic action in smooth muscles – less miosis, constipation and urinary retention
 - Rapid but short duration of action (2-3 Hrs)
 - Vagolytic effect - Tachycardia
 - Devoid of antitussive action
 - Less histamine release – safer in asthmatics
 - Better oral absorption



METHADONE

- Similar in most of pharmacological actions analgesic, respiratory depression etc.
- Highly bound to plasma protein 80 to 90%
- Metabolized in liver by – demethylation and cyclization
- Excreted in urine
- Slow action and – abuse potential is low
- **Used as substitution therapy** as opioid dependence
- Codeine is used as substitution in Methadone addiction



TRAMADOL

- Centrally acting analgesic
- Very low action on opioid receptors
- Other mechanisms involved in analgesic action – 5-HT and NE reuptake inhibition – spinal inhibition of pain
- Effective both orally and IV (100mg = 10 mg Morphine)
- Side effects are similar to Morphine but less prominent
- Well tolerated and low abuse potential
- Only partially reversed by Naloxone



OPIOID ANTAGONISTS

1. Pure antagonists: Naloxone, Naltrexone and Nalmefene
 - Affinity for all receptors (μ , δ and κ)
 - Can displace opioids bound to α -receptors
 - No action on Normal person but reverses poisoning and withdrawal symptoms in addicts
2. Mixed Agonist-antagonists: Nalorphine, Pentazocine, Butorphanol and Nalbuphine
3. Partial/weak μ agonist and κ antagonist: Buprenorphine



NALOXONE

- Competitive antagonist of all types of opioid receptors
- But, blocks μ -receptors at much lower dose
- All symptoms of Morphine action are antagonized – respiratory stimulation
- At higher doses 4-10 mg: antagonizes actions of Nalorphine and Pentazocine –
- Withdrawal symptoms: 0.4 mg doses – Morphine and 4-5 mg doses – Nalorphine and Pentazocine



BUPRENORPHINE

- Synthetic thebaine congener and **highly lipid soluble**
- **Selective μ -agonist analgesic**
 - **20-30 times more affinity than Morphine**
- Pharmacological effects are similar to **Morphine**
- **Has ceiling effect in analgesic and respiratory depression**
- **Lower tolerance and physical dependence than Morphine.**



LOPERAMIDE & DIPHENOXYLATE

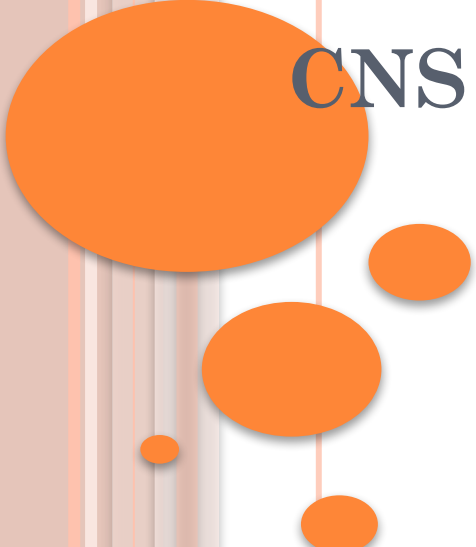
- Piperidine derivatives
- Approved for **treatment of diarrhoeas**
- **It slows GI motility**
- Also may **act by decreasing secretions**
- **Loperamide** has very less central effects **because of the activity of P-glycoprotein**



PENTAZOCINE

- **KOR agonist and a weak antagonist/partial agonist at MOR receptors**
- **Effects similar to morphine – analgesia, sedation, respiratory depression.**
- **At high doses – dysphoric & psychotomimetic effects**
- **Tachycardia & increase in BP**
- **Can precipitate withdrawal in morphine dependent patients**





CNS STIMULANTS

Molecular basis of CNS stimulation

- Balance b/n inhibitory and excitatory neurons
 - Potentiation or enhancement of excitatory neurotransmission
e.g. Amphetamine
 - Depression or antagonism of inhibitory transmission
e.g. Strychnine



CLASSIFICATION OF CNS STIMULANTS

I. Analeptics

- Respiratory stimulants
- Convulsants

II. Psychomotor stimulants

- Sympathomimetics /adrenergic CNS stimulants
- Methylxanthines

III. Psychotomimetics / hallucinogens



I. ANALEPTICS

- Diverse chemical classes
- Majority can be absorbed orally
- Effect terminated through metabolism
 - Short $t_{1/2}$
- Common mechanism of Action
 - Alteration of Cl^- conductance
 - Convulsions is the primary effect
- Used as experimental tools but have limited clinical uses



Respiratory stimulant

Doxapram

Convulsants

Pentylenetetrazole (PTZ)

Picrotoxin

Bicuculline

Strychnine



Doxapram

- Has a bigger margin of safety b/n respiratory stimulation & convulsions
- At low dose more selective for respiratory center than other analeptics
- Occasionally used as an i.v. infusion in patients with acute respiratory failure
- May be used to abolish episodes of apnoea for premature infants who don't respond for theophylline
- Counteract post-operative respiratory depression
- Associated with nausea, coughing & restlessness w/c limit its usefulness



Pentylentetrazole (PTZ)

- Precise mechanism is not known
- Antiepileptic drugs w/c inhibit PTZ-induced convulsions imply their effectiveness against absence seizures
- Used to test potential anticonvulsant drugs in laboratory animals
.....has no clinical use
- Used for screening of **latent epileptics** as it can precipitate the typical EEG pattern of absence seizures in susceptible patients
.....????



Picrotoxin

- Blocks the action of GABA_A on Cl⁻ channels but not competitively
- Obtained from the fishberry
- The plant's name reflects the native practice of incapacitating fish
- Like bicuculline, it causes convulsions & has no clinical uses
- **Diazepam** facilitate GABAergic transmission
 - **Antidote** for picrotoxin poisoning



Bicuculline

- Resembles strychnine in its effects with different mechanism of action i.e. Strychnine block glycine receptor while bicuculline block GABA receptor
- It is competitive GABA_A antagonist (block the receptor) but not affect GABA_B receptor
- The main effects are on the brain rather than the spinal cord
- Useful experimental tool for studying GABA-mediated transmission..... has no clinical uses



Strychnine

- Potent convulsant
- Tonics containing strychnine “banned”
- Acts particularly throughout spinal cord, causing violent extensor spasms that are triggered by minor sensory stimuli, the head being thrown back & the face fixed
- Mechanism of action : blocking glycine receptors
- Treatment of strychnine poisoning is similar to status epilepticus



- The action superficially resembles that of tetanus toxin produced by the *C.tetani* w/c blocks release of glycine from inhibitory interneurons
- This is very similar to the action of botulinum toxin w/c is produced by *C. genus* & causes paralysis by blocking Ach release
- In small doses, strychnine causes a measurable improvement in visual & auditory acuity



II. PSYCHOMOTOR STIMULANTS

1. Amphetamine & related compounds

(Eg. dexamphetamine, methylamphetamine, methylphenidate, fenfluramine)

2. Methylxantines

(Caffeine, theophylline)



1. Amphetamine & related compounds

- Are indirect sympathomimetic agents
 - Displace stored catecholamines from the presynaptic adrenergic nerve ending
 - Inhibit reuptake of catecholamines
- Additionally
 - Direct agonist of DA & 5-HT receptors
 - May inhibit MAO

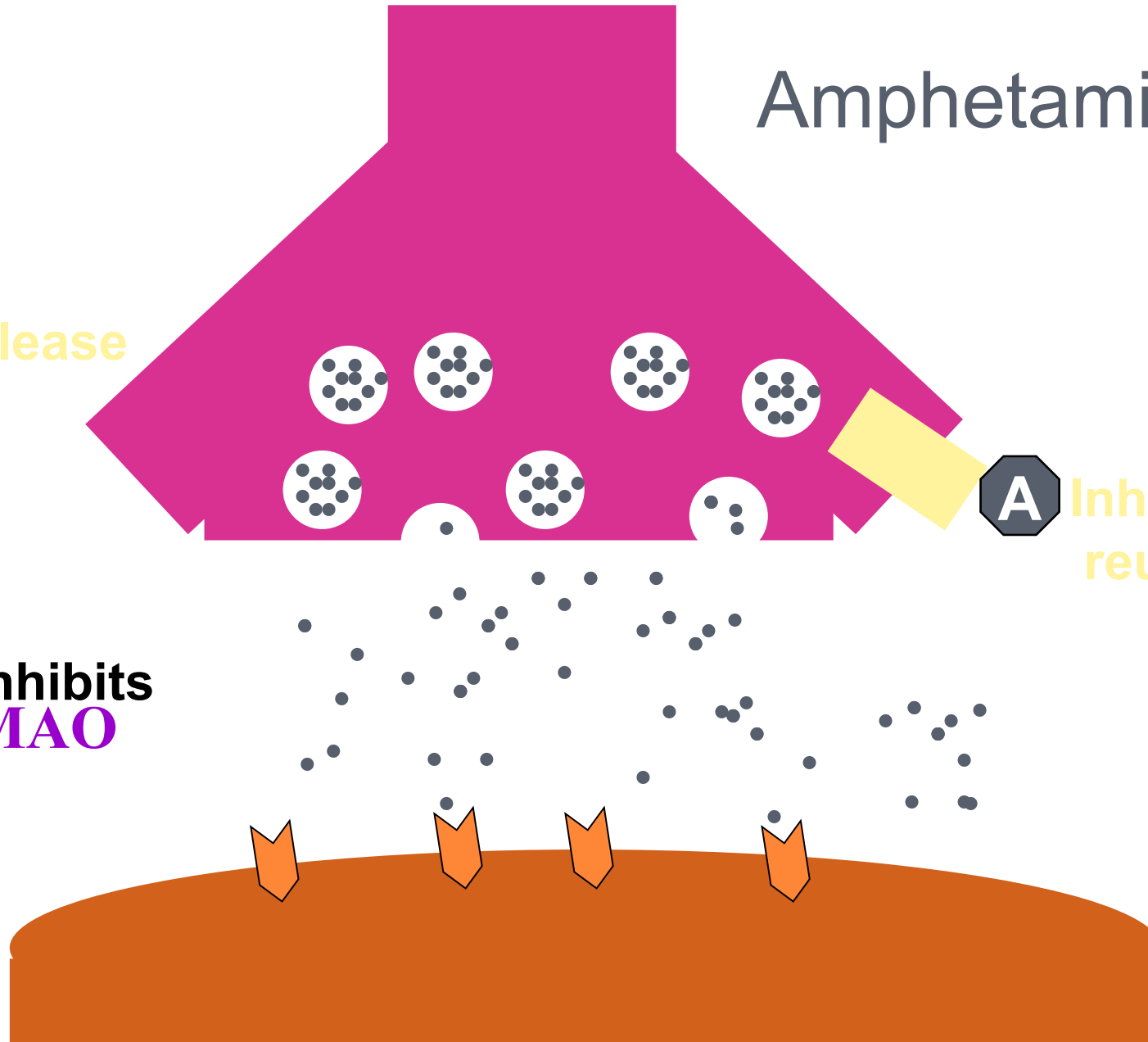


Amphetamine (A)

A release

A Inhibit
reuptake

A inhibits
MAO



PHARMACOLOGICAL ACTIONS

- Wakefulness/alertness, mood elevation, ↓fatigue, ↑ed ability to concentrate, ↑ motor & speech activity
 - locomotor stimulation, stereotyped behaviour, euphoria & excitement
- Stimulate the respiratory center especially when it has been depressed by drugs
- Anorexia / lose of appetite by inhibiting hypothalamic feeding center
 - with continued use this effect wears off in a few days & food intake may return to normal



Amphetamine & related compounds

Therapeutic uses

- For attention deficit-hyperactivity disorder (ADHD): Hyperkinesias
 - Drug used : methylphenidate
 - at doses lower than those causing euphoria & other side effects
- For narcolepsy - amphetamine / methylphenidate
- As appetite suppressants for obesity: Fenfluramine
 - Now a days it is abandoned



Amphetamine & related compounds

Adverse Effects

- Euphoria, tremor, irritability, insomnia, convulsion (at higher doses), hyperthermia & coma
- Cardiac stimulation leads to headache, palpitations, cardiac arrhythmias, anginal pain
- Hypertension
- weight loss, Psychotic reaction
- Psychic dependence, tolerance & physical dependence



Amphetamine & related compounds

- **MDMA** (**m**ethylen**e**dioxy**m**ethamphethamine) caused sudden deaths even after a single/moderate dose
- Condition resembling heatstroke, associated with muscle damage & renal failure
- Inappropriate secretion of ADH, leading to thirst, over-hydration & hyponatraemia ('water intoxication')
- Cerebral haemorrhage after amphetamine use, possibly the result of acutely raised blood pressure



Drug interactions

- Tricyclic antidepressant
- Antihypertensive agents
- Foods with high tyramine content



Amphetamine & related compounds

Misuse of amphetamines

- By soldiers, military pilots & others to increase performance & to remain alert under extremely fatiguing conditions
- By students vaguely as a means of helping to concentrate before & during exam
 - As the improvement caused by reduction of fatigue can be offset by the mistakes of overconfidence
- By athletes to ↑ performance w/c end up with deterioration



Drugs used by athletes as CNS stimulant & officially prohibited

- Ephedrine & methylephedrine
- Amphetamines & its derivatives
- Cocaine
- Nikethamide, amiphenazole, strychnine, caffeine
- Other CNS stimulants

”Dope test” for athletics



2. METHYLXANTINES

- Caffeine in coffee, tea, cocoa
- Theophylline in tea & cocoa
- Theobromine in cocoa



Mechanisms of Action

- Increase cyclic nucleotide concentration
- Blocks adenosine receptors
- Alters intracellular Ca^{2+} distribution



Pharmacological activity/adverse Effects

- Low doses: 50-250mg/caffeine
 - $\uparrow_{ed}$ mental alertness & $\downarrow_{ed}$ drowsiness
 - Lessen fatigue
- Larger doses: 250-600mg/Caffeine
 - Irritability, restlessness, tremor, insomnia, headache, palpitations & GI upset
- Very large doses: > 1000 mg
 - Overt excitement, delirium, clonic seizures



Cardiovascular System

- Chronotropic & inotropic (low doses)
- Arrhythmias at higher doses
- Peripheral vasodilation, hypotension & cardiac arrest (rapid i.v. theophylline)
- Relaxes vascular smooth muscle (Theophylline >> Caffeine)
- Diuresis (Theophylline > Caffeine)
- Xanthines shorten clotting time by increasing tissue prothrombin & factor V



THERAPEUTIC USES

- Migraine headaches
- OTC preparations
 - Theophylline: Prophylaxis for chronic asthma
 - Bronchodilator for relief of asthmatic symptoms
 - Respiratory Stimulant



ADVERSE EFFECTS

- Stimulate gastric secretions
- Dehydration-vomiting & diuresis (theophylline)
- Allergic reaction (aminophylline)
- Psychic dependence (Caffeine)



III. PSYCHOTOMIMETICS / HALLUCINOGENS

The main types

- LSD (lysergic acid diethylamide), psilocybin & mescaline
 - Actions related to 5-HT & catecholamine
- MDMA (methylenedioxymethamphetamine)
'ecstasy'
- Phencyclidine

